Theorem 11: upper bound for the cut distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem11_CutUpperBound` formalises Theorem 11 (p. 10), the cut part of Theorem 2, and Appendix E.2: the multiplicative rounding via thin matchings (E.2.1: Lemmas 36, 37, Theorem 38), the additive rounding (E.2.2: Lemmas 40–45, 47, 48, Corollary 46, Theorem 39), the Talagrand contraction lemma (Lemma 43), and the final proof (E.2.3).

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.theorem11_of_thin`: assuming Theorem 35 [HOG26] (`ThinPermutationTheorem`), there is a universal $C > 0$ such that for all graphs G, H of the same order, with $f = \delta_{\square}^{\mathrm{OT}}(G, H)$,

$$f \le \delta_{\square}(G, H) \le C\, f \ln(e/f) \ln\bigl(e \ln(e/f)\bigr).$$

`GsimOT.theorem11_power_lt_one_of_thin`: for every $\alpha < 1$ there is $C_\alpha > 0$ with $\delta_{\square}(G, H) \le C_\alpha\, \delta_{\square}^{\mathrm{OT}}(G, H)^{\alpha}$; `GsimOT.theorem2_cut_of_thin`: the same with $\alpha = 1 - \varepsilon$ (Theorem 2). The `_of_rackeShah` versions assume only [RS14].

Correspondence

Paper	Page	Statement	Lean declaration(s) and location

Theorem 11	10	$\delta_{\square}^{\mathrm{OT}} \le \delta_{\square} \le C\, \delta_{\square}^{\mathrm{OT}} \ln \frac{e}{\delta_{\square}^{\mathrm{OT}}} \ln\bigl(e \ln \frac{e}{\delta_{\square}^{\mathrm{OT}}}\bigr)$ (assuming Thm. 35)	`GsimOT.theorem11_of_thin`
			`Theorem11_CutUpperBound/Theorem11.lean:593`
			`GsimOT.theorem11`
			`Theorem11_CutUpperBound/Theorem11.lean:517`
			`GsimOT.theorem11_of_rackeShah`
			`Theorem11_CutUpperBound/FromThinMatching.lean:28`

Paper	Page	Statement	Lean declaration(s) and location

Theorem 11 (power bound), E.2.3	10, 44	$\delta_\square \le C_\alpha(\delta_\square^{\mathrm{OT}})^\alpha$ for every $\alpha < 1$ (also the $0 < \alpha < 1$ versions)	`GsimOT.theorem11_power_lt_one_of_thin`
			`Theorem11_CutUpperBound/Theorem11.lean:603`
			`GsimOT.theorem11_power_lt_one`
			`Theorem11_CutUpperBound/Theorem11.lean:550`
			`GsimOT.theorem11_power_lt_one_of_rackeShah`
			`Theorem11_CutUpperBound/FromThinMatching.lean:47`
			`GsimOT.theorem11_power_of_thin`
			`Theorem11_CutUpperBound/Theorem11.lean:612`
			`GsimOT.theorem11_power`
			`Theorem11_CutUpperBound/Theorem11.lean:579`

Theorem 2 (cut part)	3	$\delta_\square \le C_\varepsilon(\delta_\square^{\mathrm{OT}})^{1-\varepsilon}$ for every $\varepsilon > 0$	`GsimOT.theorem2_cut_of_thin`
			`Theorem11_CutUpperBound/Theorem11.lean:622`
			`GsimOT.theorem2_cut_of_rackeShah`
			`Theorem11_CutUpperBound/FromThinMatching.lean:56`

Theorem 35 [HOG26]	32	Thin permutations exist with $\alpha \le C_{\mathrm{thin}} \ln(2n) \ln\ln(4n)$ — **explicit hypothesis**	`GsimOT.ThinPermutationTheorem`
			`Theorem11_CutUpperBound/Definitions.lean:40`
			`GsimOT.IsThin`
			`Theorem11_CutUpperBound/Definitions.lean:30`

Theorem 35 from [HOG26, Thm. 1.4]	32	Derived from the `ThinMatching` library, assuming only [RS14]	`GsimOT.thinPermutationTheorem_of_rackeShah` `Theorem11_CutUpperBound/FromThinMatching.lean:18`

Theorem 39	36	$\zeta_P \le \zeta_Q + 36\sqrt{d_n}\, n^{-1/6}$, $d_n = \lceil 2\log n \rceil$ ($n \ge 2$; A, B symmetric with entries in $[0,1]$)	`GsimOT.theorem39`
			`Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:267`
			`GsimOT.dn`
			`Theorem11_CutUpperBound/Definitions.lean:76`

Lemma 40	36	Lifted kernels: $\|K\|_{\mathcal{L}^2(n)} \le t \Rightarrow \|K^{(1)}\|_{\square,\mu_Q}, \|K^{(2)}\|_{\square,\mu_Q} \le t$	`GsimOT.lemma40_fst`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:224`
			`GsimOT.lemma40_snd`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:258`
			`GsimOT.OpBound`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:75`
			`GsimOT.weight_sq_le`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:202`

Paper	Page	Statement	Lean declaration(s) and location

Eqs. (21), (22)	32	ζ_Q (`epsR`); bilinear forms with $[0,1]$-valued functions are bounded by the cut norm; testing on sets	`GsimOT.epsR`
			`Common/CutNorm.lean:105`
			`GsimOT.abs_bilin_le_cutNorm`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:178`
			`GsimOT.abs_cut_le_epsR`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:66`

Lemma 41	37	Close support functions $\Rightarrow$ close bilinear maxima (paper's proof: $m_T(Y,Y')$ via $h_Y(Tu')$ and $h_{Y'}(T^*u)$)	`GsimOT.lemma41`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:588`
			`GsimOT.lemma41_step`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:532`
			`GsimOT.abs_inner_le_bilMax_add`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:568`
			`GsimOT.abs_supportFn_sub_le`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:495`
			`GsimOT.supportFn`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:461`
			`GsimOT.bilMax`
			`Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:464`

Lemma 42	37	Symmetrisation (independent $X_a \in \{0,1\}$ with $\mathbb{E}X_a = \theta_a$)	`GsimOT.Lemma47.lemma42`
			`Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:394`
			`GsimOT.Lemma47.bern`
			`Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:321`

Lemma 43 [LT91, Thm. 4.12]	38	Talagrand contraction lemma (cited in the paper; **proved here**)	`GsimOT.ContractionPrinciple`
			`Theorem11_CutUpperBound/Definitions.lean:54`
			`GsimOT.contractionPrinciple`
			`Theorem11_CutUpperBound/ContractionPrinciple.lean:173`
			`GsimOT.ContractionAux.two_point`
			`Theorem11_CutUpperBound/ContractionPrinciple.lean:44`
			`GsimOT.ContractionAux.comparison`
			`Theorem11_CutUpperBound/ContractionPrinciple.lean:144`

Paper	Page	Statement	Lean declaration(s) and location

Lemma 44, Eq. (29)	38–39	Block Rademacher estimate $\le 2M(\sum_j \ell_j^2)^{1/2}$ (random representatives $I_j \in B_j$, Jensen, Lemma 43)	`GsimOT.Lemma47.lemma44`
			Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:208
			`GsimOT.Lemma47.sum_piFinset_comp`
			Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:187
			`GsimOT.Lemma47.radAvg_norm_le`
			Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:85

Proof of Lemma 45	39–40	Spectral theorem, Parseval, truncations A_t, B_t with $\|A - A_t\|_{\mathcal{L}^2(n)}, \|B - B_t\|_{\mathcal{L}^2(n)} \le t$	`GsimOT.spectral_entry`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:48
			`GsimOT.parseval_row`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:57
			`GsimOT.specTrunc`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:68
			`GsimOT.opBound_sub_specTrunc`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:101

Eq. (31)	39	$\|\zeta_Q(A, B) - \zeta_Q(A_t, B_t)\| \le 2t$	`GsimOT.eq31`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:296
			`GsimOT.cutNorm_le_of_eq_add3`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:139

Proof of Lemma 45	39–40	Vectors z_{ij} ($\|z_{ij}\| \le \sqrt{2}$), operator T_t ($\|T_t\|_{2\to 2} \le 1/t$), $\langle z_{ij}, T_t z_{k\ell}\rangle = (A_t)_{ik} - (B_t)_{j\ell}$	`GsimOT.specVec`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:700
			`GsimOT.specDiag`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:708
			`GsimOT.diagOp`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:788
			`GsimOT.norm_specVec_le`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:754
			`GsimOT.abs_specDiag_le`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:731
			`GsimOT.norm_diagOp_le`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:808
			`GsimOT.diagForm_specVec`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:713
			`GsimOT.inner_diagOp`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:799

Paper	Page	Statement	Lean declaration(s) and location

Lemma 45, Eq. (30)	39	Spectral approximation: $\lvert\zeta_Q - \max_{S,T} \lvert \dots \langle z_{ij}, T_t z_{k\ell}\rangle \dots \rvert\rvert \le 2t$	`GsimOT.lemma45`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:826
			`GsimOT.zonKernel`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:627

Eqs. (32)–(34)	40	Zonotope Z_Q, its support function h_Q (Eq. (33)), pseudometric ρ	`GsimOT.zonLin`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:336
			`GsimOT.cube`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:346
			`GsimOT.hSupp`
			Theorem11_CutUpperBound/Definitions.lean:66
			`GsimOT.supportFn_zonotope`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:616
			`GsimOT.rhoSupp`
			Theorem11_CutUpperBound/Definitions.lean:70
			`GsimOT.norm_zonLin_le`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:396
			`GsimOT.isCompact_zonotope`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:355

Eq. (35)	40	ρ is a pseudometric, convex in both arguments	`GsimOT.rhoSupp_triangle`
			Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:61
			`GsimOT.rhoSupp_convex`
			Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:77
			`GsimOT.hSupp_lin`
			Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:70

Eqs. (36), (37)	40–41	$\max_{u,u' \in Z_Q} \lvert\langle u, T_t u'\rangle\rvert$ is the maximum of Eq. (30) (attained at indicators); $\lvert\zeta_Q - \max_{Z_Q} \rvert \le 2t$	`GsimOT.eq36`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:646
			`GsimOT.eq36_expand`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:634
			`GsimOT.eq37`
			Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:860

Corollary 46	41	$\zeta_P \le \zeta_Q + 4t + \frac{2\sqrt{2}}{t}\rho(P,Q)$ for all $P, Q \in \mathrm{DS}(n)$	`GsimOT.corollary46` Theorem11_CutUpperBound/Corollary46_SpectralTruncation.lean:870

Paper	Page	Statement	Lean declaration(s) and location

Lemma 47	41	Merging two permutations: $\rho(P,(1-\theta)P_0+\theta P_1) \le 20Mn^{-1/3}$ (the proof gives $18M$)	`GsimOT.lemma47`
			`Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:961`
			`GsimOT.lemma47_sharp`
			`Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:810`

Proof of Lemma 47	42	Blocks along the cycles of τ with properties (i), (ii)	`GsimOT.Lemma47.exists_sep_cover`
			`Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:508`
			`GsimOT.Lemma47.exists_blocks`
			`Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:547`

Proof of Lemma 47, Eq. (38)	42–43	A good choice X ($\Delta(T_X) \le 16Mn^{-1/3}$); repairing T_X into a bijection	`GsimOT.Lemma47.exists_good_choice`
			`Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:680`
			`GsimOT.Lemma47.exists_equiv_eq_off`
			`Theorem11_CutUpperBound/Lemma47_MergingPermutations.lean:644`

Proof of Lemma 48, Eq. (3)	43	Birkhoff + Carathéodory: $Q=\sum_{s\le N}\lambda_s P_s$, $N\le n^2$	`GsimOT.exists_perm_decomp` `Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:183`

Lemma 48	43	Binary-tree merging: $\rho(P,Q) \le 20Md_n n^{-1/3}$	`GsimOT.lemma48`
			`Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:243`
			`GsimOT.merge_aux`
			`Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:127`
			`GsimOT.combo`
			`Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:103`

Proof of Theorem 39	43	$t=\sqrt{20d_n}\,n^{-1/6}$: $4t+80d_n n^{-1/3}/t = 8t < 36\sqrt{d_n}\,n^{-1/6}$	`GsimOT.theorem39` `Theorem11_CutUpperBound/Theorem39_AdditiveRounding.lean:267`
Eq. (23)	32	α-thin permutations: thinness for all cuts $(X\cup Y, \overline{X}\cup\overline{Y})$	`GsimOT.IsThin` `Theorem11_CutUpperBound/Definitions.lean:30`

Lemma 36	33	Thinness $\Rightarrow \sum_i \lvert x_i - y_{\pi(i)}\rvert \le \alpha \sum_{ij} Q_{ij}\lvert x_i - y_j\rvert$ (level sets, integral)	`GsimOT.lemma36`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:107`
			`GsimOT.levelInd`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:85`
			`GsimOT.levelInd_apply`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:87`
			`GsimOT.integral_levelInd`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:99`

Paper	Page	Statement	Lean declaration(s) and location
Lemma 37, Eq. (24)	33–34	Local agreement (for $y = \mathbf{1}_T$, the case used)	`GsimOT.lemma37` `Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:139`
Eq. (25)	34	$\|RK\|_\square \le \|K\|_\square$ for $R \in \mathrm{DS}(n)$ (one-dimensional form)	`GsimOT.abs_sum_mul_le_of_cut` `Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:30`
Theorem 38	34	There is P with $\zeta_P \le (1+4\alpha)\zeta_Q$, $\alpha = C_{\mathrm{thin}} \ln(2n) \ln\ln(4n)$	`GsimOT.theorem38` `Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:404`

Proof of Thm. 38, Eq. (26)	34	$\zeta_P \le (1+4\alpha)\zeta_Q$ for every α-thin P; $\zeta_P = n^{-2}\|A - PBP^\top\|_\square$	`GsimOT.eq26`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:373`
			`GsimOT.epsR_permMat_le`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:232`
			`GsimOT.sum_permMat_mul_atoms`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:218`

Proof of Thm. 38, Eq. (27)	35	$n^{-2}\|Q^\top AQ - B\|_\square \le \zeta_Q$	`GsimOT.eq27_left` `Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:257`

Proof of Thm. 38	35	$n^{-2}\|EA\|_\square \le 2\alpha\zeta_Q$ (Lemmas 36, 37, mean zero)	`GsimOT.normEA_le`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:285`
			`GsimOT.abs_sum_le_half_of_sum_eq_zero`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:54`

Proof of Thm. 38, Eqs. (28), (25)	35	$A - RAR^\top = EA + R(EA)^\top$; $\|R(EA)^\top\|_\square \le \|EA\|_\square$	`GsimOT.eq28_cut`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:340`
			`GsimOT.cut_R_EAt_le`
			`Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:311`

Proof of Thm. 38, Eq. (27)	35	$n^{-2}\|A - RAR^\top\|_\square \le 4\alpha\zeta_Q$	`GsimOT.eq27_right` `Theorem11_CutUpperBound/Theorem38_ThinRounding.lean:355`

Section 2; proof of Thm. 11	5, 44	$C_\square(P_\pi) = \|A_G^\pi - A_H\|_\square$; lower bound $\delta_\square^{\mathrm{OT}} \le \delta_\square$	`GsimOT.costCut_permMat`
			`Theorem11_CutUpperBound/Theorem11.lean:348`
			`GsimOT.deltaOTCut_le_deltaCut`
			`Theorem11_CutUpperBound/Theorem11.lean:455`

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 11	44	An optimal Q exists (compactness of DS, continuity of $C_\square$)	`GsimOT.isCompact_DS` Theorem11_CutUpperBound/Theorem11.lean:384 `GsimOT.continuous_costCut` Theorem11_CutUpperBound/Theorem11.lean:404 `GsimOT.exists_costCut_eq_dOTCut` Theorem11_CutUpperBound/Theorem11.lean:416
Eq. (39)	44	$\delta_\square \le C_1 \ln(2n) \ln\ln(4n)\, \varepsilon$, $C_1 = 1 + 4C_{\text{thin}}$, using $\ln 4 \cdot \ln\ln 8 > 1$	`GsimOT.eq39` Theorem11_CutUpperBound/Theorem11.lean:491 `GsimOT.one_le_log_mul_log_log` Theorem11_CutUpperBound/Theorem11.lean:281 `GsimOT.one_lt_log_four_mul_log_log_eight` Theorem11_CutUpperBound/Theorem11.lean:260
Proof of Thm. 11	44	Case analysis ($\varepsilon = 0$, $\varepsilon \ge \frac{1}{2}$, $n \le \varepsilon^{-12}$, $n > \varepsilon^{-12}$), $C = \max\{2, 39C_1, 113\}$	`GsimOT.main_numeric` Theorem11_CutUpperBound/Theorem11.lean:111 `GsimOT.error_term_le` Theorem11_CutUpperBound/Theorem11.lean:58 `GsimOT.dn_mul_log_two_le` Theorem11_CutUpperBound/Theorem11.lean:28 `GsimOT.log_fourteen_lt_three` Theorem11_CutUpperBound/Theorem11.lean:100
(not in the paper)		$n = 1$: $\delta_\square = 0$ (the paper assumes $n \ge 2$ in Appendix E.2)	`GsimOT.deltaCut_le_zero_of_card_eq_one` Theorem11_CutUpperBound/Theorem11.lean:473 `GsimOT.rhs_nonneg` Theorem11_CutUpperBound/Theorem11.lean:295
Proof of Thm. 11	44	$\varepsilon^{1-\alpha} \ln\frac{e}{\varepsilon} \ln(e \ln\frac{e}{\varepsilon})$ is bounded on $(0, 1]$, for every $\alpha < 1$	`GsimOT.power_numeric` Theorem11_CutUpperBound/Theorem11.lean:200 `GsimOT.power_numeric_lt_one` Theorem11_CutUpperBound/Theorem11.lean:243

Shared notions used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location

Appendix A	14	Cut norm $\|A\|_\square = \max_{S,T} \|A(S,T)\|$	`GsimOT.cutNorm`
			`Common/CutNorm.lean:32`
			`GsimOT.le_cutNorm`
			`Common/CutNorm.lean:39`
			`GsimOT.cutNorm_le`
			`Common/CutNorm.lean:44`
			`GsimOT.exists_eq_cutNorm`
			`Common/CutNorm.lean:52`
			`GsimOT.cutNorm_smul`
			`Common/CutNorm.lean:59`

Section 1, Section 2	2, 5	$B(G,H,Q)$ and $C_\square(Q) = \|B(G,H,Q)\|_\square$	`GsimOT.B`
			`Common/OTDistances.lean:58`
			`GsimOT.costCut`
			`Common/CutNorm.lean:79`

Section 1, Section 2	1–2, 5	$d_\square$, $d_\square^{\mathrm{OT}}$ (infimum) and $\delta_\square = n^{-2} d_\square$, $\delta_\square^{\mathrm{OT}} = n^{-2} d_\square^{\mathrm{OT}}$	`GsimOT.dCut`
			`Common/CutNorm.lean:82`
			`GsimOT.dOTCut`
			`Common/CutNorm.lean:85`
			`GsimOT.deltaCut`
			`Common/CutNorm.lean:88`
			`GsimOT.deltaOTCut`
			`Common/CutNorm.lean:91`

Appendix E.2, Eq. (21), E.2.2	32, 36	$\zeta_R(A,B) = \|D_{A,B}\|_{\square,\mu_R}$ with $\mu_R = R/n$; $\zeta_R(A_G, A_H) = \zeta_R = n^{-2} C_\square(R)$ for graphs	`GsimOT.epsR`
			`Common/CutNorm.lean:105`
			`GsimOT.adjMat`
			`Common/CutNorm.lean:94`
			`GsimOT.epsR_adjMat`
			`Common/CutNorm.lean:110`

Paper	Page	Statement	Lean declaration(s) and location
Section 1, Appendix B	1, 3, 15	$\kappa_{p\to q} = 1/q + 1 - 1/p$, $A_G^\pi - A_H$, $d_{p\to q}$ and $\delta_{p\to q} = n^{-\kappa_{p\to q}} d_{p\to q}$ $(s(n) = n^{\kappa_{p\to q}})$	`GsimOT.kappa` Common/AlignedDistances.lean:23 `GsimOT.alignedDiff` Common/AlignedDistances.lean:31 `GsimOT.distPQ` Common/AlignedDistances.lean:35 `GsimOT.deltaPQ` Common/AlignedDistances.lean:39

Notes on the formalisation

1. **Cited results.** Theorem 35 ([HOG26, Thm. 1.4], Haqi–Oveis Gharan; p. 32) is the only remaining assumption. It is the explicit hypothesis `ThinPermutationTheorem` of `theorem11_of_thin`; no `axiom` is added. Its statement matches [HOG26, Thm. 1.4] (bipartite case: an $O(\log n \log\log n)$-thin perfect matching, not necessarily in $\mathrm{supp}(x)$), written with an absolute constant as $C\ln(2n)\ln\ln(4n)$ as in the paper, with thinness in the cut form of Eq. (23). The Lean hypothesis gives thinness with exactly $\alpha = C_{\mathrm{thin}}\ln(2n)\ln\ln(4n)$, so the step "$\alpha'$-thin with $\alpha' \le \alpha$, hence α-thin" of the proof of Theorem 38 is built in. The contraction principle [LT91, Thm. 4.12] (Lemma 43), also cited in the paper, is *proved* (`contractionPrinciple`) via Talagrand's comparison argument. The paper's other citations used here are also proved: [vdVW96, L. 2.3.1] (Lemma 42) and Carathéodory's theorem [Sch86] with $N \le n^2$ (proof of Lemma 48).

2. **Theorem 35 from [HOG26].** [HOG26] is formalised in the separate library `ThinMatching` (see `ThinMatching/HOG26_correspondence.pdf`), assuming the tree-cut sparsifier theorem of Räcke–Shah [RS14] that [HOG26] cites. `thinPermutationTheorem_of_rackeShah` derives `ThinPermutationTheorem` from it, so `theorem11_of_rackeShah` proves Theorem 11 assuming only [RS14].

3. **Additive bound (Appendix E.2.2): what follows the paper.** Lemma 41 is proved with the paper's support-function argument (`lemma41_step`: $m_T(Z,Z) \le m_T(Z',Z) + M\eta\tau$ via $h_Z(Tu')$, then $m_T(Z',Z) \le m_T(Z',Z') + M\eta\tau$ via $h_{Z'}(T^*u)$ with the adjoint; then T and $-T$); the statement does not assume Z, Z' convex. Lemma 44 averages over independent uniform representatives $I_j \in B_j$ (`sum_piFinset_comp`), uses Jensen, Lemma 43 for fixed I and Eq. (29); the statement does not assume the blocks disjoint. Lemma 42 is proved for arbitrary θ_j. Eq. (36) is proved as an identity (`eq36`) and Corollary 46 for all $P, Q \in \mathrm{DS}(n)$. Expectations are finite weighted sums (`radAvg`, `bern`, uniform averages over `Fintype.piFinset`).

4. **Additive bound: choices of the formalisation.** (1) The standard inner product on $\mathbb{R}^n$ is used instead of $\mathcal{L}^2(n)$, with the scalings adjusted (eigenvalues λ_r/n, eigenfunctions $\sqrt{n}\,e_r$); $\|K\|_{\mathcal{L}^2(n)} \le t$ is `OpBound` K (tn) in the equivalent AM–GM form. (2) $H = \mathbb{R}^{V \sqcup W}$ (dimension $2n$) instead of $\mathbb{R}^{I_A \sqcup I_B}$; the extra coordinates of z_{ij} are zero, so the case $I_A = I_B = \emptyset$ needs no separate treatment. (3) The blocks of Lemma 47 come from a maximal set of "cut points", no two of them 1 to $L-1$ steps apart along τ, instead of the explicit cycle decomposition; they satisfy the paper's properties (i) and (ii), which is all the proof uses. Lemma 44 is applied to all blocks with the vectors $X_a u_i$ (zero on blocks with $X_a = 0$) instead of to the blocks with $X_a = 1$; the bound is the same. The w.l.o.g. step $P_0 = I$ is done by working with $\tau = \pi_0^{-1}\pi_1$ and $u_i = z_{i,\pi_0(i)}$ directly. The proof gives $18Mn^{-1/3}$ (`lemma47_sharp`), the statement $20Mn^{-1/3}$ (`lemma47`). (4) In Lemma 48 the binary tree splits the index set in halves at every node (all weights positive) instead of padding with leaves of weight zero; the induction is the paper's.

5. **Theorem 38 (Appendix E.2.1).** The proof follows the paper step by step: ζ_P as a cut norm (`epsR_permMat_le`), the split $A - PBP^\top = (A - RAR^\top) + P(Q^\top AQ - B)P^\top$ with $R = PQ^\top$, the two bounds of Eq. (27) (`eq27_left`, `eq27_right`), the bound on EA via Lemmas 36 and 37 (`normEA_le`), and Eq. (28) with Eq. (25)

(`eq28_cut`, `cut_R_EAt_le`). Matrices are written entrywise ($R(v,k) = Q(k,\pi v)$). Lemma 37 is proved for $y = \mathbf{1}_T$, the only case used. Eq. (25) is used in the one-dimensional form `abs_sum_mul_le_of_cut`. The bound for every α-thin P (Eq. (26)) is `eq26`, the existence statement is `theorem38`.

6. **Proof of Theorem 11 (Appendix E.2.3).** The optimal Q is obtained by compactness, as in the paper's "let Q be optimal". Eq. (39) uses $C_1 = 1 + 4C_{\text{thin}}$ and $\ln(2n)\ln\ln(4n) \geq \ln 4 \cdot \ln\ln 8 > 1$ for $n \geq 2$ as in the paper. The case $n = 1$, excluded in Appendix E.2, is handled separately ($\delta_\square = 0$). For $n > \varepsilon^{-12}$, the paper's monotonicity argument for $x \mapsto x^{-1/6}\sqrt{\ln x}$ is replaced by $\ln m \leq m/e$ with $m = n^{1/6}$; the constant 112 is the same. The constants are $C = \max\{2, 39C_1, 113\}$ as in the paper.

7. **Power bound.** Proved for every $\alpha < 1$ (`theorem11_power_lt_one_of_thin`), as stated in E.2.3; for $\alpha \leq \frac{1}{2}$ the bound for $\alpha = \frac{1}{2}$ is used together with $f^{1/2} \leq f^\alpha$ on $[0,1]$. The cut part of Theorem 2, $\delta_\square \leq C_\varepsilon(\delta_\square^{\mathrm{OT}})^{1-\varepsilon}$ for every $\varepsilon > 0$, is `theorem2_cut_of_thin`. (In Lean $0^\alpha = 0$ for $\alpha < 0$; the statement still holds since $\delta_\square^{\mathrm{OT}} = 0$ forces $\delta_\square = 0$.)

8. In Lean $\ln 0 = 0$ and $e/0 = 0$, so the right-hand side of Theorem 11 is 0 when $\delta_\square^{\mathrm{OT}} = 0$, matching the paper's convention.